

Periodic medical assessment of athletes

with STEPHEN TARGETT and BEN CLARSEN

To screen or not to screen, that is the question.
adapted from *Hamlet*, William Shakespeare (1564–1616)

Sports medicine clinicians commonly perform standardised health evaluations of athletes and many different organisations around the world require that athletes undergo a pre-participation evaluation to be allowed to participate in sport. However, organisations rarely provide guidance regarding the content of the required evaluation and the effectiveness of the approach in reducing the risk of injury, illness or sudden death in athletes remains controversial.

There are many different templates available, such as the Fédération Internationale de Football Association (FIFA) pre-competition medical assessment (PCMA),¹ the International Olympic Committee periodic health examination (PHE)² and the American Academy of Paediatrics pre-participation physical evaluation (PPE).³ It should be emphasised, however, that these are designed for different target groups and there is no one universal template that suits all situations.

In order to avoid any confusion in terminology in this chapter, we use the term 'periodic medical assessment' (PMA) to refer to the standardised medical assessment of athletes, without being specific to any group's recommended template.

The development of a template for the PMA of an athlete, whether elite or recreational, requires careful planning, and gives an opportunity for all of the members of the sports medicine team and coaching staff to work in collaboration. Establishing the optimum template for each situation can be achieved by considering the following questions in turn.

- Why perform the medical assessment?
- Who is being assessed?
- When and where should the assessment be performed, and by whom?
- What should be included in the assessment?
- Are there any other issues to consider?

WHY PERFORM THE MEDICAL ASSESSMENT?

The first step in planning a PMA template is to determine why you are performing the medical assessment. The primary goal of the PMA is to make participation in sport safer, but there are a number of other valid reasons for wanting to perform a PMA such as assessing the status of known injuries and illnesses, screening for risk factors, reviewing medications and supplements, baseline testing, athlete education and establishing rapport with athletes.

Identification of medical conditions that contraindicate participation in sport

There are some medical conditions that may contraindicate participation in certain sports or recreational activities; for example, uncontrolled epilepsy in motor racing, cycling or swimming; or human immunodeficiency virus (HIV) infection in an amateur boxer. In many cases, such as asthma in a scuba diver, the medical condition is not an absolute contraindication and athletic participation should be decided on a case-by-case basis. It is therefore important that the medical professional performing the PMA understands the physical and mental demands of the sport in question, as well as its regulations. Table 46.1 shows examples of medical conditions that warrant special attention or contraindicate participation in certain sports.

It is important to identify medical conditions that may be made worse by intense exercise; for example, uncontrolled type 1 diabetes, poorly controlled asthma or uncontrolled hypertension. Although such conditions are usually not absolute contraindications to sports participation, optimal control should be established prior to commencing intense training and competition.

Table 46.1 Medical conditions affecting sports participation (adapted from Bernhardt et al.⁴)

Medical condition	Sport/activity participation recommendations
Atlantoaxial instability	Medical evaluation necessary
Bleeding disorder, blood thinning medications	Medical evaluation necessary. Contact sport usually discouraged
Cardiovascular disease*	
Myocarditis/cardiomyopathy	All exercise contraindicated
Hypertension	Those with severe hypertension (>99th percentile for age plus 5 mmHg) should avoid heavy weight and power lifting, bodybuilding, strength training and sports with a high static component. Those with sustained hypertension (>95th percentile for age) need medical evaluation
Congenital heart disease	Medical evaluation necessary
Dysrhythmia	Medical evaluation necessary
Heart murmur	Medical evaluation necessary
Structural/acquired heart disease	
Hypertrophic cardiomyopathy	All exercise normally contraindicated
Coronary artery anomalies	All exercise normally contraindicated
Arrhythmogenic right ventricular cardiomyopathy	All exercise normally contraindicated
Acute rheumatic fever with carditis	All exercise normally contraindicated
Ehlers-Danlos syndrome, vascular form	All exercise normally contraindicated
Marfan syndrome	Medical evaluation necessary
Mitral valve prolapse	Medical evaluation necessary
Anthracycline use	Medical evaluation necessary
Vasculitis/vascular disease	Medical evaluation necessary
Cerebral palsy	Medical evaluation necessary
Diabetes mellitus	Participation normally allowed with proper attention to diet, blood glucose concentration, hydration and insulin therapy
Infectious diarrhoea	All sports normally contraindicated, unless symptoms are mild and the athlete is fully hydrated
Eating disorders	Medical and psychological evaluation necessary. Sports participation contraindicated when athlete is not compliant with therapy and follow-up, or where there is evidence of diminished performance or potential injury because of eating disorder
Eye conditions	
Functionally one-eyed athlete	Medical evaluation necessary. Protective eyewear may be appropriate, although in some sports (e.g. boxing, martial arts) it is not permitted
Loss of an eye	
Detached retina or family history of retinal detachment at a young age	
High myopia	
Connective tissue disorder, such as Marfan or Stickler's syndromes	
Previous intraocular eye surgery or serious eye injury	
Infectious conjunctivitis	
	Swimming contraindicated

Table 46.1 Cont.

Medical condition	Sport/activity participation recommendations
Fever	All exercise contraindicated
Gastrointestinal conditions	
Malabsorption syndromes (coeliac disease, cystic fibrosis)	Medical evaluation necessary
Short bowel syndrome or other disorders requiring specialised nutritional support including parenteral or enteral nutrition	Medical evaluation necessary
Hepatitis, infectious	Participation normally allowed in all sports where the athlete's state of health permits. Athletes should receive hepatitis B immunisation and universal precautions should be used when handling blood or body fluids with visible blood
Human immunodeficiency virus (HIV) infection	Participation normally allowed in all sports where the athlete's state of health permits. Universal precautions should be used when handling blood or body fluids with visible blood. Sports with a high risk of viral transmission (e.g. boxing and wrestling) should be avoided
Infectious mononucleosis—acute or with splenic enlargement	Moderate/intense exercise and participation in collision and contact sports contraindicated
Absence of one kidney	Medical evaluation necessary for collision, contact and limited-contact sports. Protective equipment may be available to protect the remaining kidney.
Acute liver enlargement	All exercise normally contraindicated
Chronic liver enlargement	Medical evaluation necessary, particularly for collision, contact and limited-contact sports
Malignant neoplasm	Medical evaluation necessary
Neurological disorders	
History of serious head or spine trauma or abnormality, including craniotomy, epidural bleeding, subdural haematoma, intracerebral haemorrhage, second-impact syndrome, vascular malformation and neck fracture	Medical evaluation necessary
History of simple concussion (mild traumatic brain injury), multiple simple concussions and/or complex concussion	Medical evaluation necessary. Research supports a conservative approach to concussion management, including no athletic participation while symptomatic or when deficits in judgment or cognition are detected, followed by a graduated, sequential return to full activity (see Chapter 20)
Myopathies	Medical evaluation necessary
Recurrent plexopathy (burner or stinger) and cervical cord neuropraxia with persistent defects	Medical evaluation necessary for collision, contact or limited-contact sports; regaining normal strength is an important benchmark for return to play
Seizure disorder with risk of seizure during participation	Athlete needs individual assessment for collision, contact or limited-contact sports. The following non-contact sports should be avoided: archery, shooting, swimming, weight or power lifting, strength training, or sports involving heights or high velocities (e.g. cycling, motor racing). In these sports, occurrence of a seizure during the activity may pose a risk to self or others
Organ transplant recipient (and those taking immunosuppressive medications)	Medical evaluation necessary, particularly for collision, contact and limited-contact sports

Continued

Table 46.1 Cont.

Medical condition	Sport/activity participation recommendations
Pregnancy	Medical evaluation necessary. As pregnancy progresses, modifications to usual exercise routines will be necessary. Activities with a high risk of falling or abdominal trauma, scuba diving and activities posing a risk of altitude sickness should be avoided. Postpartum, physiologic and morphologic changes of pregnancy take 4–6 weeks to return to baseline
Respiratory conditions	
Pulmonary compromise, including cystic fibrosis	Medical assessment necessary, participation normally allowed. Athletes with cystic fibrosis have an increased risk of heat illness
Asthma	Medical assessment necessary prior to scuba diving
Acute upper-respiratory infection	Upper-respiratory obstruction may affect pulmonary function. Athlete needs individual assessment for all but mild disease. If fever is present, all exercise is contraindicated
Rheumatologic diseases	
Juvenile rheumatoid arthritis, juvenile dermatomyositis, idiopathic myositis or systemic lupus erythematosus	Medical assessment necessary
Raynaud's phenomenon	Care should be taken when exposing hands and feet to cold
Sickle cell disease	Medical assessment necessary. In general, if status of the illness permits, all sports may be played; however, any sport or activity that entails overexertion, overheating, dehydration and chilling should be avoided. Participation at high altitude, especially when not acclimatised, also poses risk of sickle cell crisis.
Skin infections —including herpes simplex, molluscum contagiosum, verrucae (warts), staphylococcal and streptococcal infection (furuncle/boils, carbuncle, impetigo, methicillin-resistant <i>Staphylococcus aureus</i> /cellulitis/abscess/necrotising fasciitis), scabies, tinea	During contagious period, participation in gymnastics with mats, martial arts, wrestling or other collision, contact or limited-contact sports is contraindicated
Enlarged spleen	If the spleen is acutely enlarged, participation should be avoided because of risk of rupture. If the spleen is chronically enlarged, individual assessment is needed before collision, contact or limited-contact sports are played

*All suspected cases should be evaluated by a cardiologist. The 36th Bethesda Conference,⁵ the European Society of Cardiology consensus statements⁶ and the recommendations from the American Heart Association and American College of Cardiology⁴⁰ provide detailed recommendations for sports participation for cardiovascular conditions

Assessment of known injuries and illnesses

Athletes often have persisting or recurrent injury problems and the PMA is a good opportunity to check that they are receiving optimal treatment. Recent studies of American college athletes and professional footballers in Qatar found that 11% and 17% of athletes had a current injury at the time of the PMA, respectively.^{7, 8}

All medical personnel involved in the care of each existing injury should be listed. The current level of symptoms should be ascertained, as well as athlete's

compliance to ongoing rehabilitation or prevention exercises. It is also important to check that exercises are being performed correctly.

It is equally important to address chronic or recurrent illnesses. This involves checking for optimum treatment (e.g. measurement of blood pressure for those with hypertension, asking about breakthrough symptoms in asthma or epilepsy, checking blood glucose control for diabetics), monitoring compliance with treatment (e.g. taking of medication, inhaler technique and attending follow-up appointments) and checking whether the athlete has any questions and/or adequately understands their condition.

Review of current medications and supplements

This involves checking prescribed medications as well as 'over-the-counter' medications or supplements that an athlete is currently taking, takes occasionally or is considering taking. Athletes sometimes neglect to inform their team medical staff about any changes and these should be checked to ensure compliance with the current World Anti-Doping Agency (WADA) regulations and be updated in their medical records.

While discussing medication, it may also be a good opportunity to remind athletes of their responsibilities under the WADA code (www.wada-ama.org). In particular, it should be emphasised that they are ultimately responsible for all substances they ingest and may be required to report their whereabouts to anti-doping authorities. They should also be reminded about the potential hazards associated with taking supplements, minerals and non-prescription medications without first checking with medical staff.

Education

In addition to providing education about anti-doping, medication use and supplements, other issues may be worth discussing with athletes depending on the local circumstances. Although cardiac screening is a hot topic at present, the leading cause of death in teenagers and young adults is accidents, in particular road traffic accidents, meaning that education on drink driving, wearing seatbelts and safe driving practices should be considered.⁹ Other lifestyle practices that may have a negative effect on health and/or performance include abuse of alcohol, the use of recreational drugs or 'legal highs', smoking (tobacco or shisha) and unsafe sexual practices.

Careful consideration should be given to the best mode of delivery of these messages; for example, whether done on a one-to-one basis during the PMA or at a later date as a group lecture or interactive workshop. This will vary from situation to situation.

Other general health topics that may affect performance include basic nutrition advice and the importance of sleep and recovery, and should be discussed.

Baseline testing

The PMA can be used to perform baseline tests for a range of purposes if appropriate. Examples include:

- neuropsychological testing for athletes involved in contact sports to be used as a baseline following head injury (see Chapter 20)
- tests used in the monitoring of fatigue and recovery in athletes throughout a season, such as blood and/or saliva biochemical, hormonal or haematologic tests, aerobic or anaerobic fitness tests, psychological tests

such as profile of mood states (POMS) or tools to assess sleep and recovery (see Chapter 13)

- physical measures to be used as post-injury reference values (these can be important when making return-to-play decisions; see Chapter 19), such as:
 - muscle strength (Fig. 46.1)
 - single-leg hop tests (see Chapter 35)



(a)



(b)

Figure 46.1 Strength testing (a) Quadriceps strength testing using an isokinetic dynamometer (b) Testing hip abductor strength with a hand-held dynamometer

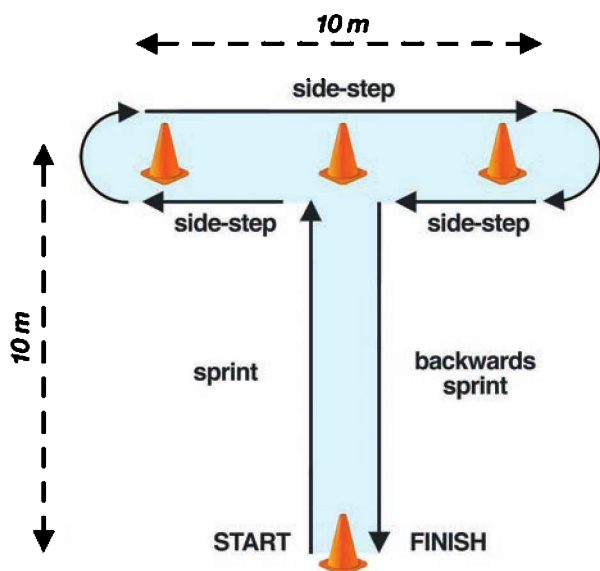


Figure 46.2 The agility T-test¹⁰

- vertical jump (counter movement jump)
- agility T-test (Fig. 46.2)¹⁰
- 30 m sprint time
- glenohumeral joint range of motion (Chapter 24).

Development of athlete rapport

The PMA is often the only time that team medical personnel have an opportunity to have a one-on-one consultation with athletes and to be proactive rather than reactive. It is an opportunity for athletes to ask questions or discuss any matters that they have been hitherto unwilling to discuss and helps in improving rapport and building the clinician-patient relationship.

In some parts of the world, where a pre-participation evaluation is required for recreational athletes, the PPE is the only contact that some athletes will have with the healthcare system and is an opportunity to establish a 'medical home' and discuss general health issues, especially if performed by a family physician.

Screening

Screening is traditionally defined as the identification of disease among apparently well persons. In 1968, Wilson and Jungner described 10 principles of early disease detection that are still relevant today when considering a screening program for athletes (see box).¹¹

Although these principles are universal, two of them are worth considering more closely in the context of sport, as they may be interpreted differently in athletes than in the general population.

Ten principles of early disease detection

1. The condition should be an important health problem.
2. There should be an accepted treatment for patients with recognised disease.
3. Facilities for diagnosis and treatment should be available.
4. There should be a recognisable latent or early symptomatic stage.
5. There should be a suitable test or examination.
6. The test should be acceptable to the population.
7. The natural history of the conditions, including development from latent to declared disease, should be adequately understood.
8. There should be an agreed policy on whom to treat as patients.
9. The cost of case-finding (including diagnosis and treatment of patients diagnosed) should be economically balanced in relation to possible expenditure on medical care as a whole.
10. Case-finding should be a continuing process and not a 'once and for all' project.

Wilson and Jungner state that the ability to be able to treat the condition adequately when discovered is perhaps the most important. For those conditions where there is no treatment or intervention that will offer a better prognosis, alerting a person to such a condition may cause actual harm. In order to know whether a treatment offers benefit, the natural history of a condition needs to be known.¹¹

First, the condition being screened for should be deemed 'important'. For athletes, particularly elite competitors, anything that negatively affects performance or leads to lost training time may be seen as very important, particularly during crucial periods of the season. In the general population, however, a 2% reduction in 5 km time or a 10% loss of strength may be insignificant.

Second, the screening program must be cost-effective. In professional sport, where athletes are often paid many thousands of dollars per week, even short periods of time loss or small performance reductions can be very costly. Prevention of minor injuries and illnesses among professional athletes may therefore be highly cost-effective.

Screening is therefore commonly considered to be a valuable process for athletes and is standard practice in many elite sports. However, the actual preventive value of mass screening programs for athletes has been questioned.^{12–15} We will examine this in greater detail in the following section where we review the most common types of athlete screening: cardiac screening, screening for unknown injuries and illnesses, and screening for risk factors for future injury.

Cardiac screening

Several high-profile cases of sudden cardiac death in athletes over recent years, such as that of Norwegian swimmer Alexander Dale Oen, as well as the case of English professional footballer Fabrice Muamba, who survived a sudden cardiac arrest after prolonged resuscitation, have stimulated a healthy debate in both the medical community and the popular press about the pros and cons of cardiac screening.

Despite all this debate, there is no universal consensus on whether or how cardiac screening should be performed. The lack of agreement is because the context in which the PMA is being performed will affect the decision on whether or not to offer cardiac screening and thus this decision will likely remain an individual one for each organisation or country.

Some of the many factors that have to be considered when deciding whether or not to offer cardiac screening and/or how to screen are the population being studied (numbers, age group, gender, prevalence of conditions predisposing to sudden cardiac death), the screening tests being used (sensitivity and specificity) and the resources available (money, equipment and access to cardiologists with expertise in conditions that predispose to sudden cardiac death).

When considering mass screening funded by the public purse, the cost-benefit of screening for conditions predisposing to sudden cardiac death versus other demands for the health dollar also needs to be considered. Many countries simply do not have the resources to be able to offer a large-scale screening program particularly if it might place barriers to people partaking in exercise, which will actually reduce public health costs.

Cardiac screening usually consists of the following elements:

- targeted history for symptoms suggestive of cardiac disease, and for personal or family history of cardiac disease
- physical examination of the cardiovascular system
- special tests such as electrocardiograph (ECG), echocardiograph or stress test.

As mentioned, there is no international consensus on which special tests should routinely be used. Most of the debate, however, centres on the routine use of an ECG as part of cardiac screening. Those against the routine use of ECG in cardiac screening point out that there is a high false positive rate, there may be undue stress associated with further investigations of 'abnormal' results, expertise is required to interpret the ECG and the costs are high. Those in favour of the routine use of ECG argue that:

- ECG is more sensitive and specific than history and examination alone; a study of college athletes showed that 30% responded positive to at least one of the American Heart Association cardiac questions, which reflects the high rate of false positives⁸
- the use of standardised ECG criteria to interpret ECGs can improve the accuracy of ECG interpretation¹⁶
- athletes feel reassured by ECG screening, not unduly stressed.^{17, 18}

Echocardiograph and stress tests are usually only routinely used as part of cardiac screening by professional sporting teams with sufficiently large budgets and access to expertise, or when further investigation is indicated by an initial screening assessment.

PRACTICE PEARL

An important part of the topic that often gets overlooked when considering cardiac screening is being adequately prepared for a sudden cardiac arrest should one occur. Even a comprehensive cardiac screening program will not identify all persons with conditions predisposing to sudden cardiac death.

High school studies in the United States clearly show the benefit of having an automated external defibrillator (AED) on site (Fig. 46.3) and that sudden cardiac arrest (SCA) is survivable with an AED program. A study published in 2013 looking prospectively at 2149 US high schools, 87% of which had an AED program, reported that 89% of the athletes (including students and older athletes) survived to hospital discharge because of prompt CPR and early defibrillation.¹⁹ Being prepared for SCA involves more than just having an AED on site.

Screening for unknown illnesses

In some situations, screening for the presence of unknown or subclinical medical conditions may be worthwhile. For example, these include any medical conditions that:

- have a negative effect on performance, such as iron deficiency anaemia or exercise-induced asthma
- predispose athletes to injury; for example, there is an association between vitamin D deficiency and stress fractures in military populations²⁰
- might be hazardous to other athletes or medical staff, such as hepatitis B.

Screening for risk factors for future injury

A comprehensive musculoskeletal examination is among the most common elements of a PMA. Typically,



(a)



(b)

Figure 46.3 A functioning automated external defibrillator (AED) is essential equipment at venues that host sporting events. (a) AED closed, showing on/off switch (b) AED open, showing simple instructions on the flap and electrodes ready to be placed on the patient's chest

COURTESY OF PROFESSOR JONATHAN DREZNER, UNIVERSITY OF WASHINGTON CENTER FOR SPORTS CARDIOLOGY, WITH PERMISSION FROM PHYSIO-CONTROL, INC.

athletes' strength and flexibility are measured and compared to normative values or pre-determined cut-off values to determine whether the athlete is at risk of future injury. Specific orthopaedic tests may also be performed in order to identify unknown existing injuries. In recent years, however, there has been a trend away from static clinical measurements towards dynamic screening of the athletes' ability to perform a range of functional tasks. A number of similar systems are currently popular, all of which aim to identify 'weak links' in strength, flexibility or motor control, which can be targeted by individualised injury prevention programs. These include the Functional Movement Screen (FMS),^{21–23} the United States Tennis Association High-Performance Profile²⁴ and the 9+ test battery developed by physiotherapists at the Swedish Sports Confederation (Fig. 46.4).²⁵

Irrespective of whether traditional clinical measurements or a qualitative approach are used, the goals of musculoskeletal screening remain the same—to identify athletes at risk of specific injuries and guide the development of individualised injury prevention training programs. Although this is a theoretically appealing approach, there are several reasons why it may not deliver the desired results in practice.

First, for most injuries very few intrinsic modifiable risk factors have been identified and screening cut-off values are often based on expert opinion or assumed associations with injury risk. For example, athletes with poor hamstring flexibility are commonly advised to stretch in order to avoid hamstring strains. However, research into the link between hamstring flexibility and injury is inconclusive, and there is no cut-off value for hamstring flexibility that can identify players at risk for future injury.^{26, 27} This is the case for the majority of physical measures currently used to screen athletes.

Second, even in cases where significant associations between risk factors and injury have been established, such as eccentric knee flexor weakness for hamstring strains²⁸ or glenohumeral joint stiffness for shoulder injuries in throwing athletes,^{29–32} intrinsic risk factors are poor predictors of future injury. This is because they typically only explain a small part of an athlete's overall injury risk and because cut-off values are likely to have low sensitivity (many false negatives) and/or low specificity (many false positives), depending on which value is used.¹⁵

Clearly, more high-quality research into injury risk factors is required, in particular into the combination of test findings in order to maximise their predictive value. It is likely that dynamic predictive models are necessary before athletes at risk of future injury can be successfully identified.³³



Figure 46.4 Examples of tests within the 9+ test battery. A full description of the tests and scoring criteria can be found in Frohm et al.²⁵

COURTESY OF A FROHM, SWEDISH SPORTS CONFEDERATION

Nevertheless, musculoskeletal screening remains a popular and potentially valuable tool, particularly in helping sports medicine personnel familiarise themselves with each athlete's physical attributes and limitations. Care should be taken, however, in telling athletes that they are not at risk of injury or that they don't need to perform injury prevention training, because they passed a particular screening test.

WHO IS BEING ASSESSED?

There are numerous factors to consider about the population being assessed that can affect the content of the PMA.

Sport and position

The incidence and type of injuries or medical issues varies widely between sports. For example, in soccer the major injuries are knee injuries (ligament and meniscal tears), ankle ligament sprains, thigh muscle strains (hamstrings and quadriceps) and groin injuries; this is different to many other sports.³⁴ In some sports such as rugby, the incidence of injuries differs between positions. A study of professional rugby union players in the United Kingdom showed that anterior cruciate ligament (ACL) injuries caused the most time loss for forwards and hamstring injuries for backs.³⁵

It is important to know about the injury and illness epidemiology for the particular athlete's sport and position as the PMA should specifically look for any risk factors associated with the important injuries or medical issues associated with each athlete's sport, and seek to address any of those that are modifiable. Some sporting bodies require athletes taking part in international competitions to undergo a medical assessment prior to competing, such as the FIFA football World Cup, the World Rugby (IRB) World Cup and the Olympics. A set template will usually be provided and should be followed.

Geographical location

In some countries, legislation dictates that some athletes require a medical assessment prior to playing sport. The Italian Government has required the screening and medical clearance of athletes since 1971 with an annual physical examination, personal and family history, and 12-lead ECG for all athletes aged 12–35 years participating in competitive sport.

Local patterns of disease, both infective and genetic, vary around the world and should be taken into account when designing a PMA. For example, in certain parts of the world there is a high prevalence of hepatitis B. The most recent guidelines from the US Preventive Task Force recommend mass population screening for hepatitis B in areas with a prevalence of over 2% in the

general population.³⁶ However, screening for hepatitis B immunity in order to be able to offer immunisation to those who are non-immune may be appropriate, even in countries with a lower prevalence rate for athletes involved in contact sports.

Extreme environmental conditions such as heat, cold and altitude can be experienced by some athletes as part of their sport when played in some geographical locations. Any previous history of complications associated with such conditions should be explored in the history when athletes are likely to encounter such environmental stresses.

Age

There is no consensus on what age to start performing PMAs. The American Academy of Paediatrics recommends annual routine health screening for all healthy children and adolescents from age 6, but there is no evidence to support starting PMAs in sporting groups at any particular age. As mentioned previously, in Italy an annual cardiac screening is mandatory for all competitive athletes from age 12. The European Society of Cardiology recommends pre-participation cardiac screening from ages 12 to 14, and Cardiac Risk in the Young (CRY) in the UK recommends screening from age 14.

Some sports have an increased risk of overuse injuries to certain growth plates (epiphysitis or apophysitis); for example, the proximal humeral epiphysis or medial humeral epicondyle epiphysis in pitchers and the distal radial epiphysis or calcaneal apophysis in gymnasts (Chapter 44). Symptoms of growth plate injuries associated with the particular sport being played should be enquired about during the PMA in young athletes who are between the ages at which these different conditions occur and a change in technique or training load may be required. Although the majority of these heal with adequate rest, there have been case reports of growth arrest of epiphyses when not treated properly.

If at all practicable, a parent or legal guardian should accompany a child to the PMA. Studies of high school students undergoing a PPE have shown that only 19–33% of athletes' responses were the same as those given by their parents or legal guardians when completing the same form.^{37, 38} If a parent is unable to attend, then the previous medical history and family history sections of the questionnaire should be completed by them prior to the consultation.

Sex

Female athletes will generally have the same basic PMA template as males playing the same sport; however, the range of conditions associated with relative energy

deficiency needs to be considered in female athletes, particularly in those who are involved with high-risk sports (those with weight categories to meet or where the aesthetic can be judged, such as gymnastics or ballet).³⁹ Several screening tools have been proposed to identify the athlete with relative energy deficiency. However, none have been validated and a high index of suspicion is required. It should be noted that, although much less common, relative energy deficiency may also occur in male athletes.

It is known that female athletes have a 2–3-fold higher incidence of ACL injury than their male counterparts and that male athletes have a higher incidence of hamstring injuries. Such differences in injury rates, when present, should be reflected in the medical assessment.

Available resources

At the end of the day, the available budget will usually dictate the content of the PMA, irrespective of the desired components. A major hurdle for organisations considering developing a publicly funded PMA cardiac screening program for large numbers of athletes is the availability of cardiologists with expertise in sports cardiology who will be needed to follow up those with cardiovascular symptoms, abnormal physical examination findings or abnormal ECGs.

In elite or professional sport, where money may be less of an issue and there are often full-time medical staff available to perform medical assessments, access to equipment (e.g. for isokinetic or VO_2max testing) or expertise (cardiologists with expertise in sports cardiology) may instead limit the content of the PMA.

WHEN TO PERFORM A PMA

The PMA is typically performed in the pre-season period. This is often the first time that the medical staff get to meet any new members of the squad who have joined in the off-season and is a good opportunity to develop a working relationship with new players.

It is a good idea for medical staff to engage coaching staff and management in the development and planning of the PMA so that they understand the goals (and limitations) and are encouraged to think about injury prevention. This makes them more likely to allow adequate time for medical staff to perform the medicals and be supportive of the delivery of any recommendations that might come out of the PMA, such as individual or team-wide injury prevention programs.

The value of engaging coaching staff has been clearly shown with the ACL injury prevention program in elite female handball players in Norway.⁴⁰ Despite an

intervention study showing that a series of rehabilitation exercises done routinely as a part of team training significantly reduced the incidence of ACL injury, most teams stopped performing the exercises once the study was completed and the study personnel driving the injury prevention exercises were removed from the clubs; ACL rates consequently rose in subsequent seasons. A reduction of injury rate was seen after the introduction of education sessions for coaches and a sport-specific website on injury prevention.⁴⁰

Ideally, a shorter modified PMA is also performed at the end of the season to review any injuries that might have occurred during the season (including ones that may not have been reported to medical staff), to possibly repeat some of the baseline musculoskeletal and/or laboratory measurements that were performed earlier in the season and to formulate a treatment or rehabilitation program over the off-season.

No research has conclusively shown that more frequent PMAs reduce the incidence of death or injury and recommendations on frequency vary. The most common recommendation is that a comprehensive assessment be performed every 2-3 years with a shorter, annual questionnaire and targeted examination in between. However, practice varies depending on the context and available resources.

WHAT TO INCLUDE IN THE TEMPLATE

The basic structure of the medical template will consist of history, examination and investigations. A detailed history is an important part of the assessment because the most consistent risk factor for injury across injury types is a previous history of injury. Defining the goals of the PMA (which, as referred to earlier, can vary depending on the stage of the season) and taking into account the unique factors associated with the population in question (including the available resources) will allow you to select the appropriate tools for the task.

The PPE monograph from the American Academy of Paediatrics is an excellent resource for anyone designing a PMA. It contains sections on the history and a systems-based examination, and has an example of a PMA form.³

The temptation when designing a PMA template is to include too much, particularly in the musculoskeletal examination. For example, taking reliable measurements of strength or joint range of movement is time consuming and may be a waste of time if no action is going to be taken for 'abnormal' results (what is 'abnormal' is often not even considered), if the data are not going to be used as a baseline post injury or if the data are not being used as part of a research project.

OTHER ISSUES TO CONSIDER

Consent

The PMA form should contain a consent section that the athlete should sign covering the following points, if appropriate:

- the athlete consents to undergo a medical assessment to assess their fitness to safely participate in sport
- the athlete understands that clearance may not be given if any significant medical issues or injuries are discovered
- the athlete certifies that their answers to all PMA questions are true to the best of their knowledge
- the athlete agrees to inform the medical professional if their medical situation changes significantly after the completion of the PMA
- the athlete consents to the use of data collected as part of the PMA for research purposes as long as the results do not in any way identify the athlete
- a parent or legal guardian should sign the consent form for athletes under the age of consent (18 in most countries).

Clearance or restriction from play

A very small percentage of athletes undergoing a PMA will be diagnosed with a medical condition that is associated with an increased risk of sudden cardiac death. In such cases, a cardiologist with the appropriate experience should review the athlete. The assessment of risk, however, is not a precise science and may differ between physicians. In order to make the decisions more consistent, guidelines have been produced for participation in sport with different cardiovascular conditions; however, the guidelines do differ in places and the decision on whether to allow the athlete to participate should be individualised once all the relevant information is available.^{6, 41}

If clearance to participate is not given, the athlete may ask to sign a 'risk release' acknowledging that they have been fully informed of the risks and assume the risk of participating against medical advice; however, this may not be appropriate in all cases. It is important that documentation of any risk release is completed correctly and some experts recommend that the athlete and/or parents write in their own words their understanding of the risks of continued participation to prove that they have been fully informed of the risks. A decision to withhold clearance from participation may be subject to legal challenge. The success of a challenge will depend on the circumstances of the individual case and the local legislation, which varies widely from country to country.

Who should perform the PMA?

Although some parts of the PMA are often performed by other health professionals (e.g. nurses may take history and vital signs, physiotherapists may perform musculoskeletal assessments), the overall responsibility to review all the PMA components and make recommendations on clearance and follow-up should lie with a medical doctor who has the clinical training to deal with the wide variety of issues that might arise as part of the assessment.

Pre-employment medical assessment

Many professional athletes will be required to undergo a medical before their employment contract becomes valid. The principles for creating a template for the medical assessment are the same as for any PMA; however, there are some differences that are worth considering.

The biggest difference is the doctor-patient relationship. The person performing the medical assessment is often the athlete's team physician, with whom they have a normal doctor-patient relationship. When completing a new contract medical assessment, the doctor is acting on behalf of a third party (the club/employer). The fact that the athlete understands this difference and that that athlete agrees to the doctor providing a report on the athlete's ability to play sport to the employer using medical information from the assessment needs to be reflected in the consent form.

Even though the consent form requires athletes to promise to provide honest answers, there is a temptation for athletes to be economical with the truth (or even to lie) during a new contract medical, especially if the contract is worth large amounts of money. The physician performing the assessment should bear this in mind and if possible attempt to see the previous medical records. For some sports, injury information as well as matches and minutes played is freely available on the internet and is a valuable source of information and should be obtained if possible.

For high-value contracts, some clubs routinely ask for multiple magnetic resonance imaging (MRI) scans as part of the new contract medical (e.g. both knees, ankles, groin and lumbar spine in soccer players). The value of this approach is unknown as an MRI scan often picks up asymptomatic findings, such as a chondral defect in a knee that may have little or no practical prognostic value when taken in the context of a 1-year contract, but can be useful as a baseline when repeating imaging at a later date. Clubs can also use this information to negotiate a clause in the contract to protect themselves against prolonged future time loss from identified abnormalities.

In addition to the usual goals of a standard PMA, the role of the medical team performing a new contract medical is to provide the potential employer with an

estimated level of risk that any identified medical and/or injury issues might preclude the athlete from being able to perform at their best for the duration of their contract. Some studies show a link between previous injury and future playing ability or length of career, such as the reduced performance of running backs in National Football League (NFL) after ACL reconstruction,⁴² or the effects of previous injury history noted at the NFL Combine medical assessment on future career length.⁴³ The Combine study also showed, however, that athlete talent was a big factor that also had to be taken into account.

The prediction of future injury risk is complex and likely depends on a number of different factors in addition to injury history. This was highlighted by a recent pilot study at the Aspetar Orthopedic and Sports Medicine Hospital in Qatar (as yet unpublished), where a group of experienced sports physicians were asked to predict the risk of future injury for a series of case scenarios. The responses showed markedly different estimations of risk (from high to low) for almost all cases.

Once the risk has been conveyed to the potential employer, which is best done verbally so that any questions can be answered, it is up to the employer to decide whether they offer the athlete a contract. This can be quite a complex decision and involves many different factors such as depth of squad in the athlete's chosen position, the experience and talent of the athlete, the contract cost and duration, and the sport or position played.

Insurance medical assessment

Many athletes take out income protection insurance and will require a medical certificate to be completed by a doctor. In the majority of occasions, the insurance company will provide a form with the required template. If the athlete does not provide a form, they should be instructed to contact their insurance company to provide one as the requirements can vary between different companies.

It is important that the form is completed accurately as any incorrect answers can render the insurance void and may place the doctor at risk of legal action if there is any suggestion that there was any attempt to be economical with the truth. If the athlete is putting undue pressure on the team doctor to withhold certain information, the medical might be better performed by an independent doctor.

Action points from the PMA

Any action points identified as a result of the PMA should be discussed with the athlete and written on a discharge summary, which is then given to the athlete. This should also clearly indicate who is responsible for initiating the various action points and when follow-up is required (if possible, a follow-up appointment should be given at the

same time). If appropriate, the discharge summary could also be sent to any relevant stakeholders, such as team medical staff or the athlete's personal physician. However, it is important to get the athlete's consent prior to sharing any health information.

For some athletes who spend a lot of time travelling and/or are members of several teams, there can be some challenges around distribution of the results of the PMA to the various medical teams looking after them to avoid any unnecessary duplication of tests or to refer to when required. There is no one solution that will suit all athletes and this should be dealt with on a case-by-case basis.

SUMMARY

The PMA can have many uses that have been outlined in this chapter. There is no one PMA that suits all situations. Care should be taken when designing a PMA to ensure that it meets the required goals for your particular situation and also avoids performing unnecessary (and possibly time consuming and expensive) examinations or tests. Communication of the results to the athlete and an action plan to address any identified issues are important to maximise the benefits of the assessment.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

